

ORIGINAL ARTICLE

The point-of-care use of a facial phenotyping tool in the genetics clinic: Enhancing diagnosis and education with machine learning

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Abstract

Computer-assisted pattern recognition platforms, such as Face2Gene® (F2G), can facilitate the diagnosis of children with rare genetic syndromes by comparing a patient's features to known genetic diagnoses. Our work designed, implemented, and evaluated an innovative model of care in clinical genetics in a heterogeneous and multicultural patient population that utilized this facial phenotyping software at the point-of-care. We assessed the performance of F2G by comparing the suggested diagnoses to the patient's confirmed molecular diagnosis. Providers' overall experiences with the technology and trainees' educational experiences were assessed with questionnaires. We achieved an overall diagnostic yield of 57%. This increased to 82% when cases diagnosed with syndromes not recognized by F2G were removed. The mean rank of a confirmed diagnosis in the top 10 was 2.3 (CI 1.5–3.2) and the mean gestalt score 37.6%. The most commonly suggested diagnoses were Noonan syndrome, mucopolysaccharidosis, and 22q11.2 deletion syndrome. Our qualitative assessment revealed that clinicians and trainees saw value using the tool in practice. Overall, this work helped to implement an innovative patient care delivery model in clinical genetics that utilizes a facial phenotyping tool at the point-of-care. Our data suggest that F2G has utility in the genetics clinic as a clinical decision support tool in diverse populations, with a majority of patients having their eventual diagnosis listed in the top 10 suggested syndromes based on a photograph alone. It shows promise for further integration into clinical care and medical education, and we advocate for its continued use, adoption and refinement along with transparent and accountable industrial partnerships.

KEYWORDS

facial phenotyping, machine-learning, dysmorphology, models of care

1 | INTRODUCTION

The clinical skill of interpreting facial dysmorphology to formulate accurate differential diagnoses is both difficult to master and dependent on previous training and experience. After a history and physical

examination, workup of a child with a suspected genetic syndrome may involve a review of the medical literature or comparison of a patient's physical features to other known genetic syndromes. Current educational resources include pictures that may not reflect a multicultural population. Training is also complicated by limited exposure to

genetic conditions that by definition are rare ($< 5:10,000$) or ultra-rare ($< 1:100,000$) (CADTH, 2016; Richter et al., 2018). The evolving and expanded phenotypes of many of these conditions and the discovery of novel conditions further adds to the difficulties in mastering this art. Due to advances in computer-assisted pattern recognition and machine learning (ML), there is currently a platform to aid clinicians in the diagnosis of individuals with rare genetic syndromes. Face2Gene® (F2G) (FDNA Inc) uses ML to compute how similar data points from a patient's facial photograph are to a known genetic syndrome. In doing so, it creates a heat-map, gestalt, or similarity score (0–1) and generates a differential diagnosis listing the top 30 syndrome matches. F2G can allow the user to independently ascertain if their overall assessment of dysmorphic features is accurate and help build a library of differential diagnoses to advance this essential skill.

The goal of this project was to create, implement, and evaluate an innovative patient care delivery model using the leading clinical genetics facial phenotyping tool, F2G, into the current workflow in the Division of Clinical and Metabolic Genetics at The Hospital for Sick Children. We hypothesized that, with automation, faster and more accurate diagnoses would be made. This would in turn lead to more accurate use of targeted genetic testing. In a similar light, there is currently a paucity of published photographs on the racial and/or ethnic variation of many common dysmorphic syndromes. As such, many diagnoses are missed in populations under-represented in the medical literature (Lumaka et al., 2017). Importantly, the accuracy of F2G is constantly improving as more solved cases are added to its database. Thus, the use of this tool in our multicultural patient population has the potential to improve patient diagnosis in a novel and important way. It could specifically improve the facial descriptions for patients from diverse populations, who are traditionally underrepresented in educational resources for genetic professionals. Since the software also provides a broad differential diagnosis and places an element of objectivity in the analysis of facial dysmorphic features, and hosts a databank of images and teaching tools, another objective was to enhance the education of trainees via case presentation at a weekly case conference.

There is recent evidence of the usefulness of facial phenotyping tools in certain select patient populations (e.g., 22q11 deletion syndrome, Cornelia de Lange and others) (Basel-Vanagaite et al., 2016; Kruszka et al., 2017; Mishima et al., 2019; Pode-Shakked et al., 2020) but its utility at the point-of-care has only been examined retrospectively in small patient samples or case reports (Arora et al., 2020; Elmas & Gogus, 2020; Gurovich et al., 2019; Mishima et al., 2019; Narayanan et al., 2019; Zarate et al., 2019). Our study contributes an important look into this new intersection of ML and medicine in a multicultural pediatrics genetic clinic in Canada and the first exploration of providers' and trainees' experience with the tool, both as a diagnostic tool and a teaching aid.

2 | METHODS

Since this study's main goal was to evaluate and improve the quality of healthcare and teaching, it was initially approved as a Quality Improvement (QI) initiative at The Hospital for Sick Children, with

retrospective results review approved by the Research Ethics Board (# 1000066122). Informed consent was obtained from all participants and/or their substitute decision-makers.

2.1 | Facial analysis

Beginning in August 2018, all patients seen at the Division of Clinical and Metabolic Genetics for consultation, in whom a genetic diagnosis was suspected and who consented to standard-of-care clinical photographs were asked to have their data analyzed via F2G. Any appropriate genetic testing was ordered by the clinician, irrespective of the decision to participate. In all cases, the DeepGestalt ML algorithm within F2G was used to analyze the facial data points from a patient's clinical photograph. The algorithm analyzes the mathematical representations of the patient's facial features to suggest possible diagnoses along with a corresponding gestalt or similarity score (0–1). Phenotypic data could be added by the physician, but this did not impact the gestalt score and we chose not to look at this feature. We created three cohorts. Cohort 1: Children who had whole-exome sequencing (WES) performed and a clear genetic diagnosis achieved, Cohort 2: Children diagnosed with a genetic condition who belong to a population underrepresented in the medical literature and Cohort 3: Cases that were brought to weekly case conference, which may represent a diagnostic dilemma. Diagnostic yield was measured by assessing if F2G listed the eventually confirmed genetic diagnosis in its first 10 suggested syndromes. If the confirmed diagnosis was suggested by the software, the mean rank and gestalt score were calculated. A one-sample test of a binomial proportion used to find confidence intervals. Time to clinical and molecular diagnoses were measured from the time of initial consultation to the time of chart documentation of each respective diagnosis, if they were achieved. Information on race and/or ethnicity was extracted from the chart notes.

2.2 | Clinician and trainee experience

Providers' experience with the technology were assessed with a questionnaire, looking at factors like acceptability and overall experience with the tool. This was sent after 1 year of implementation. We similarly assessed genetics trainees' educational experience with a second questionnaire, focusing on comfort with dysmorphology and the formulation of a differential diagnosis before and after the implementation of this tool in weekly education case presentation rounds.

3 | RESULTS

3.1 | Facial analysis

Between August 2018 and September 2019, 298 patients consented for their clinical photographs to be uploaded for use with F2G. At the time of this analysis, 226 patients were excluded as either no

molecular diagnosis was achieved, or confirmatory genetic testing was still pending. In the remaining 72 cases, a molecular diagnosis was confirmed (Figure 1).

Overall, the molecularly confirmed eventual diagnosis appeared in the “top 10” suggested syndromes by F2G in 41/72 cases, providing a diagnostic yield of 57% (95% CI 45.5–68.4%) (Table 1). The most common test used to achieve a diagnosis were gene panels, but microarrays and WES were also used. In Cohort 1, where an exome was the test modality selected by the clinician, 8 of 15 patients' diagnoses appeared in the top 10, giving a diagnostic yield of 53.3% (95% CI 28.1 to 78.6%). In Cohort 2, where the patient was identified as being from a racial or ethnic minority, 31 of 46 patients' diagnoses appeared in the top 10, giving a diagnostic yield of 67.4% (95% CI 53.8–80.9%). Finally, in Cohort 3, where cases were presented at Genetics case conference and may represent a diagnostic dilemma, 15 of 23, or 65% (95% CI 45.8 to 84.7) had their eventual diagnosis in the top 10 suggested by F2G (Table 1). We saw a statistically significant difference in diagnostic yield based on race/ethnicity, with a greater diagnostic yield in underrepresented populations compared to individuals of presumed White/European descent ($p = 0.025$). We also saw an increased diagnostic yield in cases where targeted gene-based testing was pursued compared with chromosomal microarray or exome testing that was not hypothesis-driven ($p = 0.025$). However, these data are likely impacted by the lack of composite images for chromosomal conditions in general, such that the algorithm could not be trained to recognize (Table 2).

If a molecular diagnosis was achieved, the mean rank in the top 10 was 2.3 (CI 1.5–3.2) and the mean gestalt score was 0.376 (SD 0.297) but the latter appeared consistent with a bimodal distribution (Figure 2a), whereas the background gestalt score of all diagnoses in the top 10 or top 30 formed a skewed distribution below a gestalt score of 0.5 (50%)(Figure 2b,c). We saw no statistically significant difference in diagnostic yield by age (Figure 3a) or sex (Figure 3b).

Interestingly, while designed for prospective use as a clinical decision support tool, we found 42 patients' photos which were inputted after a diagnosis was made. Of these previously diagnosed individuals, F2G achieved a diagnostic accuracy of 56%, which was similar to the prospective group and overall yield (Figure 3c).

Regardless of the eventually confirmed diagnosis, the most common suggested diagnosis was Noonan syndrome, followed by mucopolysaccharidosis and 22q11.2 deletion syndrome (Figure 4a,b). In 52 cases, Noonan syndrome was listed as a possible diagnosis when it was not the true diagnosis. These top 10 background lists differed by ethnicity (Figure 4c,d). The most common co-suggested differential diagnoses are depicted in Figure S1.

3.2 | Clinician and trainee experience

While all 25 clinical staff including trainees, genetic counselors, and staff in our division were invited to participate, there were a total of 15 users in the department over the course of the year-long project.



FIGURE 1 CONSORT flow diagram for cases included in this study

TABLE 1 Patient characteristics and diagnostic yield by cohort

Variable	Groups	Group counts (%)	Diagnosis in top 10	Diagnosis not in top 10	p-value
	All	72 (100%)	41/72 (56.9%)	31/72 (43.1%)	n/a
Cohort 1: Diagnostic method	Exome	15/72 (20.8%)	8/15 (53.3%)	7/15 (46.7%)	0.025
	Chromosomal	23/72 (31.9%)	7/23 (30.4%)	16/23 (69.6%)	
	Gene based	34/72 (47.2%)	26/34 (76.5%)	8/34 (23.5%)	
Cohort 2: Race/ethnicity	Presumed White/ European descent	26/72 (36.1%)	10/26 (38.5%)	16/26 (61.5%)	0.03
	Underrepresented patients/Non-White	46/72 (63.9%)	31/46 (67.4%)	15/46 (32.6%)	
Cohort 3: Case conference	Presented	49/72 (68.1%)	15/23 (65.2%)	8/23 (34.8%)	0.47
	Not presented	23/72 (31.9%)	26/49 (53.1%)	23/49 (46.9%)	
Sex	Female	30/72, (41.67%)	19/30 (63.3%)	11/30 (36.7%)	0.49
	Male	42/72, (58.33%)	22/42 (52.4%)	20/42 (47.6%)	
Age (years – mean ± SD)	All	4.72 ± 5.57; 72	4.71 ± 5.66	4.72 ± 5.54	n/a
			(41/72; 56.9%)	(31/72; 43.1%)	
			0.01 to 28	0 to 16	

TABLE 2 Availability of composite image in cases stratified by diagnostic method and race and/or ethnicity

Outcome			No composite image	Has composite image
Clinical diagnosis in top 10	Diagnostic method	Chromosomal	0	7
		Exome	0	8
		Gene-based	0	26
	Race/ethnicity	Presumed White/European Descent	0	10
		Underrepresented patients/Non-White	0	31
Clinical diagnosis not in top 10	Diagnostic method	Chromosomal	13	3
		Exome	6	1
		Gene-based	3	5
	Race/ethnicity	Presumed White/European Descent	13	3
		Underrepresented patients/Non-White	9	6

Possibly reflecting this uptake, there were 10 provider responses to the questionnaire. The responses indicated that staff saw the value in taking pictures of patients with facial dysmorphism for diagnosis and were open to using F2G in their practice. They also noted that whilst families were receptive to taking pictures for the medical records there was some more reluctance noted in taking pictures for use with F2G. Finally, there were staff concerns noted over perceived accuracy, ease of use, the consenting process, and the privacy of patient data (Figure 5).

There were a total of 23 case presentations using F2G, in our weekly case conference rounds. A 60% response rate to our questionnaire before and after implementation was achieved. Trainees appreciated the use in the software as an educational tool to help with formulating differential diagnoses, enabling the inclusion of patient photographs, and allowing retrospective review of cases presented in rounds (Table 3).

4 | DISCUSSION

Overall, F2G listed the eventually confirmed diagnosis in its top 10 suggested syndromes in 57% of the time. This is considerably lower than the maximum diagnostic rate of 90% quoted in the literature (Gurovich et al., 2019). The rarity of some of the syndromes we see in our tertiary care genetics service that may not yet have a composite image available for the algorithm to be trained upon, and the natural exclusion of some more obvious syndromes by the nature of our referral process, might account for some of this decreased yield. When cases without a composite image and not recognized by F2G were removed from our analysis, the top 10 accuracy in this group was 82%, approaching previously published analyses (Gurovich et al., 2019). Importantly, our lower yield is still clinically relevant in that a majority of patients have their eventual diagnosis listed in the top

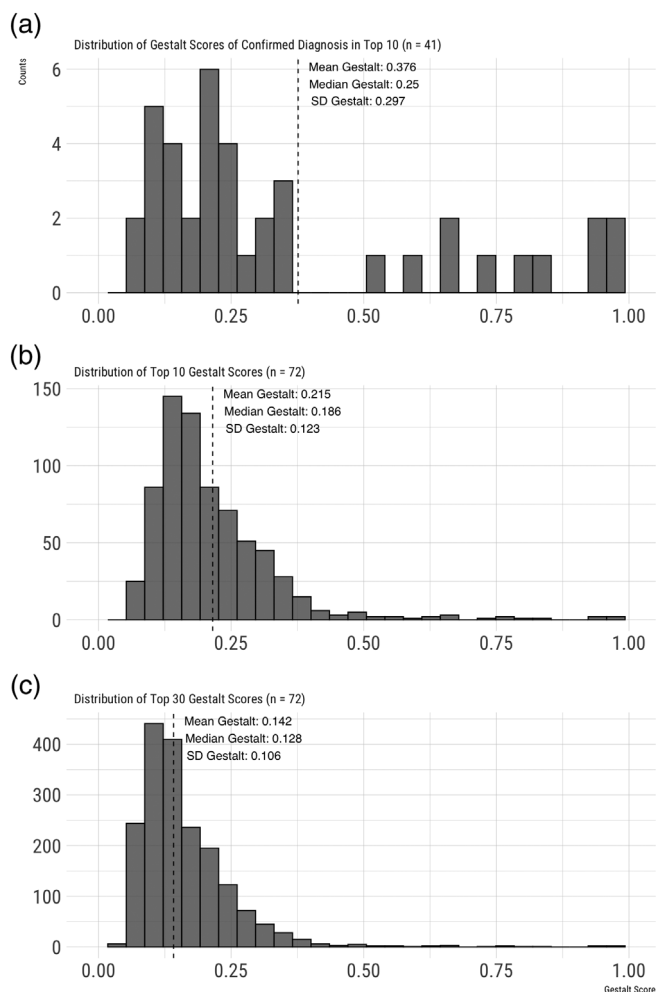


FIGURE 2 Distribution of gestalt scores in confirmed diagnoses (a), top 10 suggestions (b) and top 30 suggestions (c)

10 suggested syndromes at the initial encounter based on a photograph alone. This suggests a utility of paying close attention to this initial list, with an insight into the background “noise” of commonly suggested syndromes.

While Noonan syndrome was the most commonly suggested syndrome, it was only diagnosed three times. When it was the true diagnosis, a handful of other RASopathies were suggested alongside Noonan syndrome (Costello, LEOPARD, Noonan Syndrome Like Disorder with Loose Anagen Hair) suggesting a heightened awareness to the fact it may represent a real diagnosis when accompanied by certain conditions in the same pathway. Interestingly, if cardiofaciocutaneous syndrome and NF-1 made the list, as these were more commonly suggested overall, their presence did not reflect a true Noonan syndrome diagnosis (Figure S1). When not the true diagnosis, the background distribution of gestalt scores tended to be less than 0.5 (50%) and the mean gestalt score was 0.376 (37.6%) in confirmed diagnoses (Figure 2). Overall, we suggest using a gestalt score of 0.5 is the better proxy cutoff for likely real findings as this is above the 0.376 mean gestalt in confirmed diagnoses but does not overlap with the background distribution of gestalt scores.

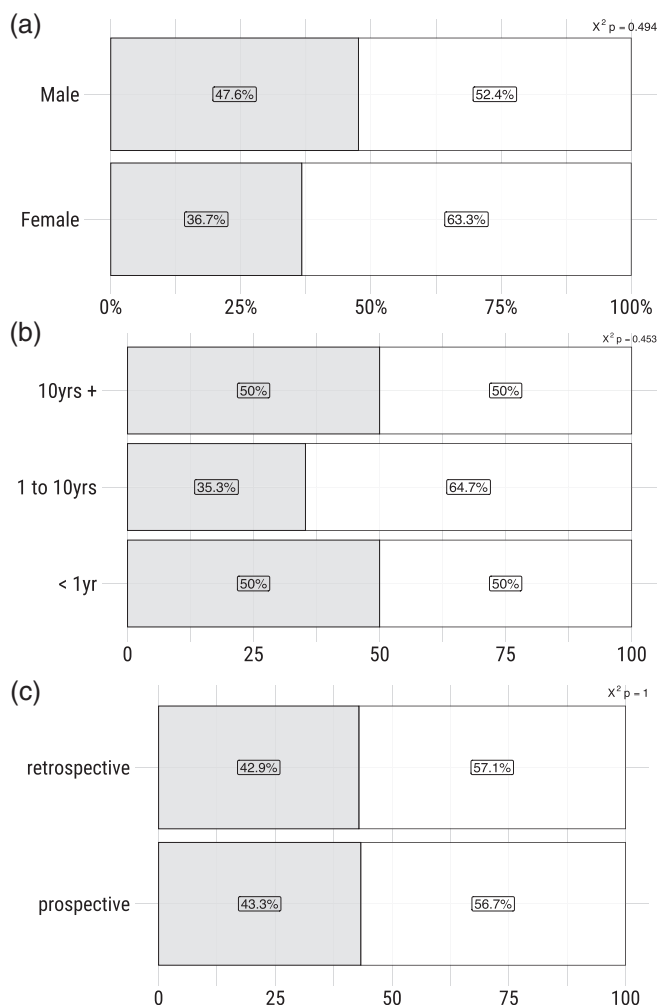


FIGURE 3 Diagnostic yield by gender (a), age (b), and retrospective versus prospective cases (c). If the clinical diagnosis appears in the top 10 syndromes suggested by Face2Gene the bars are shaded grey

Interestingly, we saw an increased diagnostic accuracy in our population of underrepresented patients, contrary to our initial hypothesis. This might be reflective of some of the factors above, namely sample bias for seeing these patients in the first place, along with the fact that many of our presumed White/European descent patients in this sample had chromosomal conditions which typically were not recognized by the algorithm, did not have a composite image and therefore could not achieve top 10 by gestalt score alone. Some users did use other phenotypic data entry, which may increase diagnostic yield further, but we did not account for this in our analysis.

Some more recent work has focused on the use of F2G to define and establish facial gestalts for specific genetic syndromes (Arora et al., 2020; Pascolini, Agolini, Fleischer, Gulotta, et al., 2020; Pascolini, Agolini, Fleischer, Pierantoni, et al., 2020; Rethanavelu et al., 2020). Anecdotally, we noticed F2G was quite good at identifying some syndromes with confidence, perhaps speaking to users uploading confirmed cases of these conditions to help train the algorithm. For example, we experienced a few high gestalt score

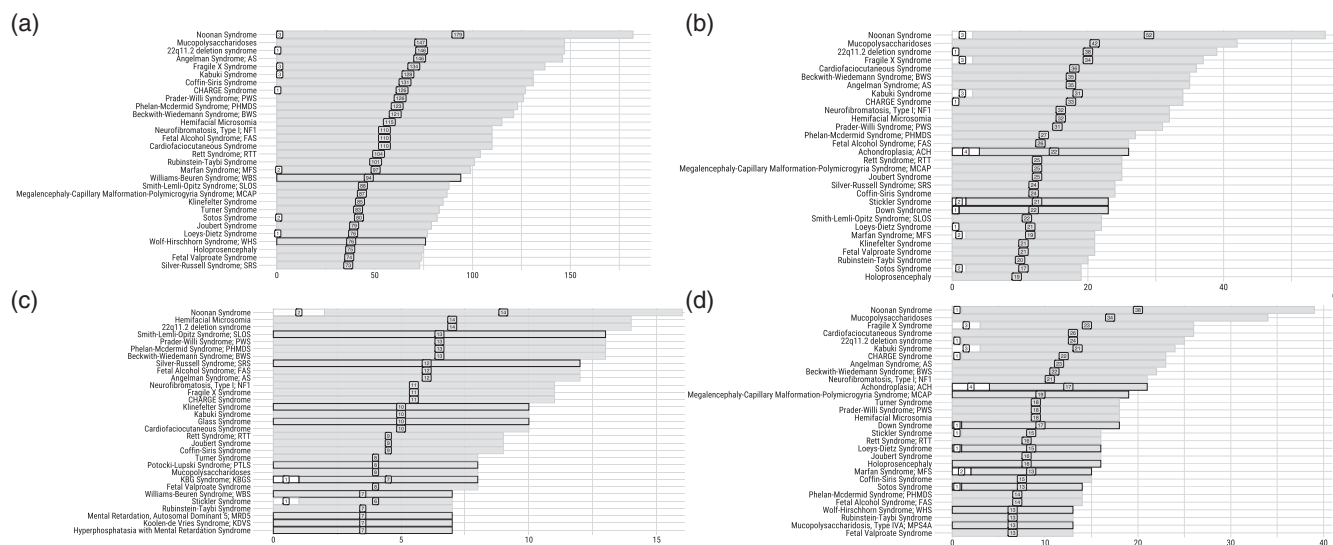


FIGURE 4 Most common suggested diagnoses in all 300 cases uploaded (a), 72 cases where diagnosis confirmed (b), Individuals of presumed White/European descent (c) and Underrepresented/Non-White patients (d). Clear bars indicate suggestions that were the correct diagnosis and shaded bars indicate suggestions that were not the correct diagnosis. Unique syndromes to a cohort have a dark box surrounding their bar chart

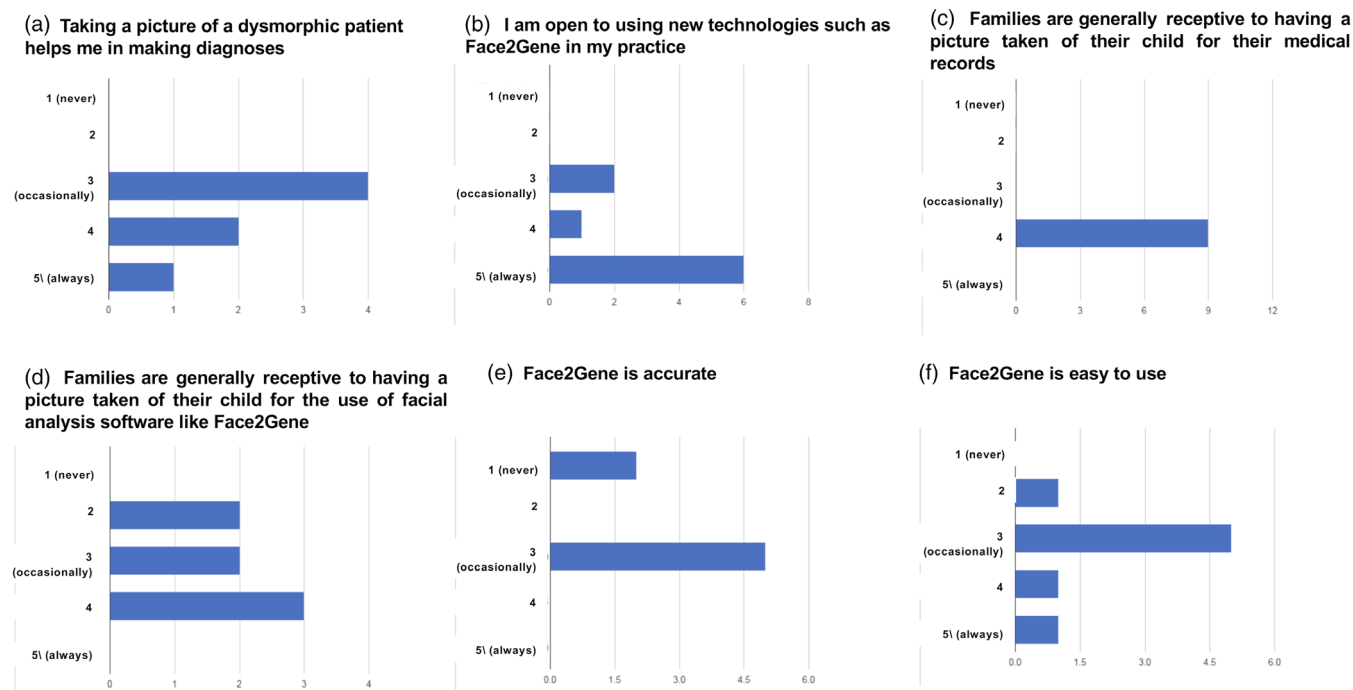


FIGURE 5 Staff questionnaire Likert scales. Questions were posed with a range of options from 1 (never) to 5 (always) [Color figure can be viewed at wileyonlinelibrary.com]

suggestions of rare single-gene disorders, presumably due to the prior training of the algorithm for that particular condition. On at least one occasion, this actually changed our clinical approach from WES to a targeted panel. In a real-world setting, though, rare syndromes might not have composite images to be recognized, which was likely the case for many of our chromosomal diagnoses and may account for the decreased yield in that group. We had one case of Trisomy 13 in a

9-year-old, for which survival is typically poor and images later in life not as common that the algorithm did not recognize. Other factors also seemed to influence the differential diagnosis. For example, smiling patients tended to have Angelman listed, as published photographs of this syndrome typically portray this character trait. We had one photograph whose quality was not optimal, but we changed the contrast and had different syndromes suggested, suggesting an

TABLE 3 Trainee questionnaire

Question	Before implementation	After implementation	Difference in Likert score
Presenters develop a differential based on dysmorphic features in the case they are presenting in genometic rounds	3.592	3.668	+0.076
Presenters include patient photographs in their presentations in genometic rounds	3.883	3.917	+0.034
I review cases discussed historically in genometic rounds for my own learning	2.527	3.003	+0.476

Note: Average Likert score before and after implementation of Face2Gene in genometic rounds.

important technical element that requires standardization. Given the distribution of genetic disease across age, sex, and ethnicity, training bias will influence the accuracy of F2G across these domains. Nevertheless, this work allows for an understanding of the common output and strengths and limitations of F2G. Transparency on the part of the developer of these issues has value in being able to confidently interpret results in clinical practice.

As with any new technology, and especially so with those involving pediatric patients, large repositories of data, including facial data points and known genetic diagnoses, ethical issues abound that require thoughtful reflection. As these are so important, we decided to report these insights separately (see McCradden et al., 2020). This separate ethics piece hopes to provide important guiding steps toward the thoughtful integration of ML into patient care. Our qualitative analysis on staff experience seemed to reflect a willingness to appreciate the potential usefulness of the software in aiding diagnosis but also revealed many reservations about the accuracy, informed consent, confidentiality, and patient uptake that require further exploration. Additionally, we encountered multiple barriers in the implementation of this work, mostly related to poor uptake in our division and the ethics of implementation which represent other areas for further work.

Finally, we also explored the use of F2G as an educational tool to present cases to current genetics trainees in the division. Overall, our qualitative assessment revealed that the presentation of photographs via this platform aided in formulating differential diagnoses and enabling trainees to revisit cases for educational purposes. It has the capacity to serve as the hub for soon to be deployed competency-based education in clinical genetics. In essence, specific and expert feedback is provided with each case and represents a component of deliberate practice.

5 | CONCLUSIONS

This work helped to create and implement an innovative patient care delivery model in clinical genetics that utilizes a facial phenotyping tool at the point-of-care. While we report a diagnostic accuracy less than previously published work, we still find a 57% diagnostic accuracy impressive in all-comers. We also report the first use of the tool prospectively to aid in clinical decision-making in a real-time fashion, suggesting a role for this tool in routine care in clinical genetics.

There are potential future applications as its diagnostic accuracy improves in screening, such as for more streamlined care and faster and more accurate diagnoses. Future uses for triaging, disease discovery and variant interpretation warrant further work, but so, too, do ongoing evaluations of the opinions of patients, families, and the public.

Platforms like F2G can be seen as new tools in a clinical geneticist's (or another clinician's) "toolbox." It has shown promise in its potential for teaching, expanding the differential diagnosis, and increasing clinician confidence in a particular diagnosis. Of course, we echo previous work that such tools be used to complement a physician's clinical judgment, and not replace them. We thus advocate for its continued use, adoption, and refinement along with transparent and accountable industrial partnerships and ongoing ethical reflection.

Lessons learned from the point-of-care use of a facial phenotyping tool

- A majority of patients at our academic center had their eventual diagnosis listed in the top 10 suggested syndromes based on a photograph alone.
- The overall diagnostic yield in our clinical cohort was 57%. This increased to 82% when cases diagnosed with syndromes not recognized by Face2Gene were removed from our analysis.
- Some syndromes appear repetitively as suggested syndromes in the top 10. We suggest a gestalt above 0.5 (50%) could indicate more likely a real diagnosis.
- Some syndromes were identified with confidence while others were not recognized at all. Technical elements may also be important and transparency on the part of the developer has value in being able to confidently interpret results in clinical practice.
- Face2Gene has shown promise in its potential for teaching, expanding the differential diagnosis, and increasing clinician confidence in a particular diagnosis.
- We encountered numerous ethical issues related to facial phenotyping that requires further exploration.

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CONFLICT OF INTERESTS

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AUTHOR CONTRIBUTIONS

All authors were involved in the study conception, design, data acquisition, analysis, and interpretation of results. Lauren Chad and Ashish Marwaha drafted the manuscript and all authors revised and approved the version to be published.

DATA AVAILABILITY STATEMENT

Due to the sensitive nature of facial information coupled with genetic information, and the nature of our consent process, our data is not publicly available, but data that support the findings of this study may be available on request.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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